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PI Name:	Lysakowski, Anna Ph.D.		
Project Title:	An Ultrastructural Study of Development in Vestibular Hair Cells, and their Cellular Responses to Hypergravity		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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	The broad long-term objective of this research is to understand sensory processing in the vestibular system. Specific aims are: (1) to provide a detailed ultrastructural analysis of regional variations in cellular and synaptic architecture in the utricular macula of the developing and adult mammal, and (2) to study the influence of hypergravity on synaptic activity, synaptic plasticity and aging in the rat utricular macular hair cell. In previous work, we have described regional variations in synaptic innervation of the crista. Similar studies are underway for the utricular macula in the developing, adult, and aged mammal. These studies will provide baseline data for future flight studies. In addition to this, we are examining the effect of hypergravity conditions upon certain features of synaptic innervation, namely plasticity of synaptic innervation and the influence of hypergravity upon calyceal invaginations (one possible indicator of synaptic activity), and the cellular structure of hair cells, particularly in aged animals. Some of the cellular and synaptic features that vary regionally are likely to be a physiological response to stimulation We are defining the anatomical limits of the		

Task Description:	striolar region of the utricular and saccular maculae and presented results on the saccular macula this past year. In these studies, we have both qualitatively and quantitatively delineated the borders of the striola using a molecular marker, the calcium-binding protein calretinin. This marker stains pure calyx afferents, which are restricted to the striola. Using this marker, we have done a comparative survey of several of the rodents commonly used in inner ear studies: mouse, rat, gerbil, guinea pig, chinchilla and squirrel. Knowing the precise borders and the proportions of calyx afferents contained in the striola will help us in our hypergravity studies of synaptic activity in hair cells in both normal adult and aged animals. This study has been submitted for publication. In addition to the studies on the striolar morphometry, we also presented preliminary results on our hypergravity experiments. Multiple samples were taken from each rat utricular macula. Each sample spanned the entire sensory epithelium and included material from all three regions: striola, juxtastriola, and medial and lateral extrastriola. Dissector counts of synaptic ribbons and calyceal invaginations (another possible indicator of synaptic activity) were made as described previously in a study of the chinchilla crista ampullaris (Lysakowski & Goldberg, 1997). Preliminary results from utricular macula indicate that there are slightly lower numbers of synaptic ribbons per hair cell in the rat and that these numbers do not vary by region, but that calyceal invaginations are more numerous in the striola, compared to the extrastriola. Synaptic ribbons in type II hair cells are larger and occur in clusters more often in the striola compared to the extrastriola, similar to our results in the crista. Type II hair cells also have more ribbons per hair cell than do type I hair cells (about 15 vs 8). Preliminary results of experiments in four samples from spun adult animals indicate that there were more synaptic ribbons and calyceal invaginations in the excited striola than in the inhibited striola, in type II hair cells. A similar effect is now seen in the extrastriola for type I and type II hair cells. We are continuing to analyze more samples.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	With the background information on the normal synaptic innervation patterns that we now have, we can determine how this pattern is altered by linear acceleration, the appropriate stimulus for the utriculus. This project, which focuses on how cells respond to both acute and long-term variations in gravity, is directly relevant to one of the stated goals of NASA's Space Biology Program for Cellular and Molecular Biology. This goal is "to elucidate the mechanisms by which cells... respond to gravitational perturbations" (from NASA's Research Announcement: Research Opportunities in Space Life Sciences, June, 1998). Up till now, the scientific literature offers no answer to the question of whether or not there is a critical period for vestibular sensory development. Our preliminary data suggest that there is. This project will provide baseline data for future studies on the influence of hypergravity and microgravity on animal growth and development. It is directly relevant to one of the stated goals of NASA's Space Biology Program for Developmental Biology. This goal is "to determine the mechanisms by which gravity induces normal development and function" (from NASA's Research Announcement: Research Opportunities in Space Life Sciences, June, 1998). A second developmental, and cellular, change that we are investigating as an unexpected result of our studies is that of altered cytoskeletal changes in the hair cells of aged animals due to hypergravity conditions. This project is directly relevant to another of the stated goals of NASA's Space Biology Program for Cellular and Molecular Biology, which is to "understand how direct or indirect effects of altered gravitational forces impact...cytoskeletal structure and function" (from NASA's Research Announcement: Research Opportunities in Space Life Sciences, June, 1998).
Task Progress:	Objective 1: To extend our ultrastructural analysis of regional variations in cellular and synaptic architecture to the utricular macula of the adult rat and the developing mouse. We have recently published two papers (Desai et al., 2004a,b) comprising a comparative study of the architecture of the rodent (mammalian) peripheral vestibular sensory apparatus in the Journal of Neurophysiology. These papers are relevant to the both the biomedical model systems and the human countermeasures program because they show how similar, and how different, are data from closely related species, and how these features scale with size up to human and larger species. Several interesting findings relevant to the saccular and macular otolith organs, the detectors of linear acceleration and gravity, emerged from this study. The size and shape of the striola was measured with several metrics in three samples from each of six rodent species. Some of our results from the mouse and rat are shown. Average values for the sensory epithelia in these adult animals will serve as baseline data for comparison to spaceflight animals, in which the overall size or shape of the organ might be affected, thereby producing erroneous density measurements (e.g., hair cells, afferents or otoliths per unit area). We have also precisely measured the striolar region (the region containing the most sensitive vestibular afferents), so that there are points of reference for future studies of its morphology and physiology. One unexpected finding was that, compared to the other 4 species, only the mouse and rat had type II hair cells with strong calretinin staining in the adult. This is a feature of embryonic tissue (Zheng and Gao, 1997), but had not previously been described in detail in the adult (Sans et al., 2001), where calretinin staining shifts to the calyx afferent population. Some type II hair cells, in cross section, appeared to be immature hair cells (Rüsch et al., 1998), especially in the distal extrastriola, suggesting that the adult rat and mouse continue to add hair cells to their sensory epithelium, a feature that also has implications for spaceflight studies. One conclusion that can be drawn from these two studies is that even the normally-pigmented Long-Evans rat is far from an ideal animal model for study of the vestibular organs. Its cristae and otolith organs are smaller than can be predicted based upon its body weight (Desai et al., 2004b), and it has an unusual pattern of hair cell calretinin staining (70-80% of type II hair cells are labeled with calretinin), similar to that found in immature rodents where all hair cells are calretinin-labeled. During this funding period, we also completed two extensive review chapters, one on the comparative morphophysiology of the vestibular periphery in toadfish, frog, turtle and mammals (Lysakowski and Goldberg, 2004) and a second chapter on mammalian vestibular hair cells, with a similar comparative approach (Eatock and Lysakowski, in press). In the latter chapter, some of our developmental studies of potassium and sodium ion channel expression localization, done in collaboration with Dr. Ruth Anne Eatock on an NIH grant, have also been summarized (Eatock and Lysakowski, in press). Objective 2: To study the influence of hypergravity on synaptic activity, plasticity and aging in the rat utricular macula and the developing mouse. Ms. Maha Ahmad, a graduate student research assistant in the Dept. of Anatomy and Cell Biology, has devoted her research efforts full-time to this aspect of the project, which forms the basis of her Ph.D. dissertation. She has presented her results at ARO and at three annual Space Biology meetings and is now working on two full-length publications, one on the normal synaptic innervation in the rat and mouse vestibular periphery, and the other on the hypergravity results. The latter manuscript will show that the effects of hypergravity are maximal in the most sensitive hair cells and afferents. We have shown that the number of calyceal invaginations is significantly modified under conditions of controlled hypergravity. Control rats were anesthetized and placed in cages in the same room during the hypergravity spins, but were not themeselves subjected to hypergravity. We have now collected ten samples and completed our survey. Overall, synaptic ribbons did not show an increase in numbers on one side versus the

	other, while calyceal invaginations are significantly decreased on the excited side of the striola only. The results are based upon a total of 338 hair cell equivalents (123 in the invaginations data). In particular, the numbers of type II cells are very low.
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